

SWE-Bench Pro ↑

Direct 59

Argus 78

AARRI-Bench ↑

Paper 68.3

Argus 76.8

How often it needs you

longer bar is better

40.7 h

Argus

≈1 h

a chat-style agent

Share of time working

while it has a task in hand

95–98.7%

normal

45%

when the GPU broke

ARGUS RUNTIME

M

Manager

task · vertical · stage

Manager I/O

Task + review

Vertical + stage

keep · revise · replace

the Reviewer may challenge the plan; the Manager decides whose authority it touches

P

Planner

what is worth
doing next

Q2

E

Engineer

research · build · measure
may self-review low-risk work

R

Reviewer

read-only · cannot edit evidence
done-continue-blocked-replan

Q1

Q3

OPERATOR

credentials · budget
publish · irreversible
only these stop

reconsider

revision

SHARED WORKSPACE

Knowledge

Wiki · Skills

Log

Events · Decisions

Artifacts

Code · Results

LifeSupervisor · Backlog · Budget · Daemon · Memory

Math data ↑

28.0

Argus

20.8

Arbor

8.3

Claude

6.3

Codex

Upstream review

4 kernels

accepted · no blocking changes

fla-org

outside maintainers

nanoGPT ↓

Argus 79.77 s

Human 80.18 s

SOL-ExecBench

#6

7 top-3

2 wins

101 kernels

DELIVERED WORK — the same runtime, four domains, judged by tools and people it does not control

AI for INFRASTRUCTURE

GPU kernels · serving · local

flash-linear-attention

4 kernels

RWKV6 1.21× · KDA 1.29× · AttnRes 1.10× · SM100 fix
reviewed by outside maintainers; no blocking changes

FlashDA — dLLM masks on FA4

19/19

6 mask families, SM80+SM90; within 5–14% of native FA4
2.5–4.2× over Triton · under ¥80 of model spend

SGLang · MiniMax-H3

+11,263 ln

unified model integrated: 160/160 exact, 5.108× at BS8
62 GiB video model on a 24 GB laptop · 6.159× on A6000

AI for SCIENCE

materials · mathematics

MOF chemical control

92.5 / 100 / 74.5%

metal · node nuclearity · ligand family, from 3,300 requests
only 16.7% of parameters trained; 4/8×B200 run by Argus

Structure judgment — AUC

0.594 → 0.833

a simpler sampler beat the published one at equal compute,
admitted only after removing a confound from the comparison

Erdős–Gyárfás campaign

6 updates

six proof-backed frontier updates; one falsified route kept
with its evidence, so later work need not re-enter it

AI for SILICON

ACE-2 accelerator

Functional closure

13,914 / 13,914

24-layer Qwen2.5-0.5B W4A8, 18 operators, bit-exact
certified by a Reviewer that did not do the work

Mapped SKY130 synthesis

0.614 mm²

62,283 cells at 100 MHz; WNS and TNS both 0.00 ns
the certificate publishes its exclusions, not just passes

Human involvement

8 days · 6 asks

roughly one interruption every 32 hours; 2 of the 6
were the server dying, not the runtime asking